

photo-resist. Creating circuits with smaller and smaller features requires shorter and shorter wavelengths of light.

Technology node

In semiconductor manufacturing, we define the 7nm node as the technology node following the 10nm node. Node is merely a label for a group of technologies that deliver an improvement rather than a measure of half-pitch or gate length. The term node is used by the semiconductor industry to characterise major targets in their manufacturing roadmap. Transistors involve fin-shaped pieces of silicon, where each fin is about 14nm wide, 60nm to 70nm tall and 100nm or longer.

Gate-all-around FETs are the most likely approach at 7nm, with germanium as a channel option, which assumes that certain difficulties of adapting germanium for n-channel can be overcome.

SEMICONDUCTOR MATERIALS AVAILABLE

Material	Abundance	Ranking
Silicon (Si)	277,100	2
Phosphorus (P)	1000	11
Carbon	480	15
Gallium (Ga)	18	35
Germanium (Ge)	1.8	53
Arsenic (As)	1.5	55
Indium (In)	0.049	68

Currently, germanium is an excellent p-channel material but more difficult to work with for n-channel. Slow caches can recognise substantial gains over 28nm, but fast cache scarcely improves over the current generation. If gate-all-around and other technologies get perfected by, say, 2025, these may take over by 2030.

In any case, there are two basic transistor candidates at 7nm: FinFET and lateral gate-all-around nanowire FET, sometimes called lateral nanowire FET. At 5nm, the industry is leaning towards lateral nanowire FET. For 7nm FinFETs, chipmakers need to make the fins taller to boost drive current. FinFETs also have new channel materials to boost mobility.

The second option, lateral nanowire FET, is basically an evolutionary step from FinFET. It increases the gate area, so that you are more effective at turning off the device.

Nanowire FETs provide better electrostatics than FinFETs, but nanowire FET is more difficult to make in a fab. Still, FinFET and nanowire FET use many of the same process steps. One of the main differences is that nanowire FETs require complex processes on the bottom of the device. For 5nm, or even for 7nm, foundry experts are gearing up to develop further next-generation transistors; front runner among these seems to be what is called

gate-all-around transistor.

Gate-all-around FETs are similar in concept to FinFETs, except that gate material surrounds the channel region on all sides. Depending on design, gate-all-

around FETs can have two or four effective gates.

Gate-all-around FETs have been successfully characterised both theoretically and experimentally. These have also been successfully etched onto indium-gallium-arsenide (InGaAs) nanowires, which have higher electron mobility than silicon. InGaAs is a ternary alloy (chemical compound) of indium, gallium and arsenic. Indium and gallium are both from boron group (group III) of elements, while arsenic is a pnictogen (group V) element. Thus, alloys made of these chemical groups are referred to as III-V compounds.

Because these are from the same group, indium and gallium have similar roles in chemical bonding. InGaAs is regarded as an alloy of gallium-arsenide and indium-arsenide with properties intermediate between the two, depending on the proportion of gallium to indium. InGaAs is a semiconductor with applications in electronics and optoelectronics. The graph in Fig. 3 shows 90nm, 65nm, 45nm, 32nm, 22nm, 14nm and 10nm to have around two years gap in every succession.

32nm/28nm was an inflection point below which it really was difficult to scale down. Double patterning and then multiple patterning started taking place. FinFET was invented, and now we are looking at gate-all-around and other innovative transistor structures, EUV and so on, to go below 10nm. Definitely, a few of these will have long maturity curves with major production volumes. It needs clever and strategic planning for fabs to reap the benefits from these; these would become cash cows in the long run.

Fabricating transistors is one part of the process, often called front-end-of-line (FEOL) process. Back-end-of-line process is for all interconnections, and there comes

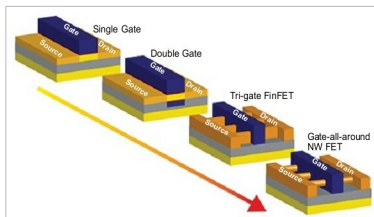


Fig. 2: Gate-all-around FET